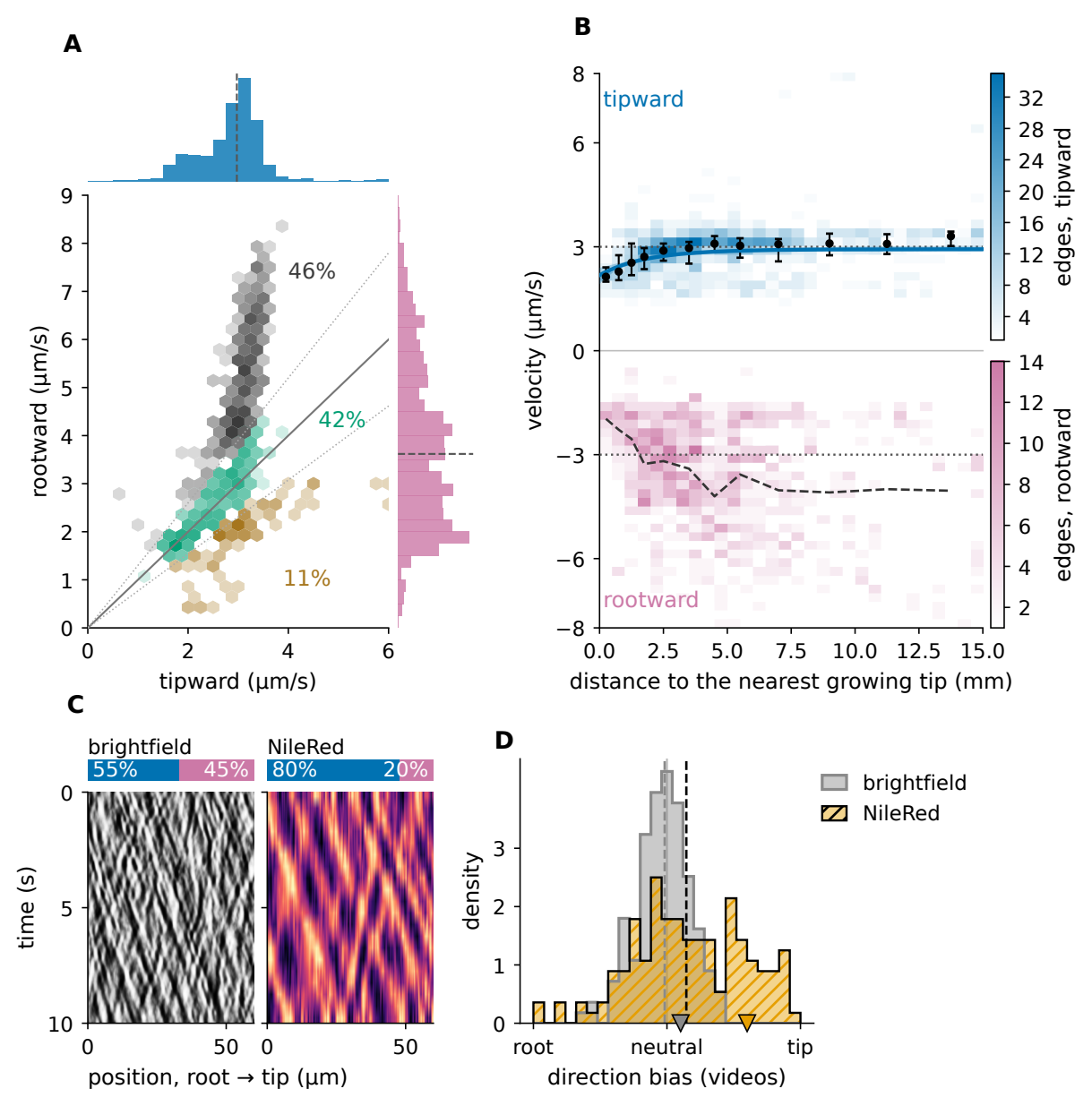

A
B
C
D
46%
42%
11%
rootward (µm/s)
tipward (µm/s)
tipward
rootward
velocity (µm/s)
edges, tipward
edges, rootward
distance to the nearest growing tip (mm)
brightfield
NileRed
55%
45%
80%
20%
time (s)
position, root → tip (µm)
density
root
neutral
tip
direction bias (videos)